

Non-Uniform Hypothesis in Deductive Databases with Uncertainty

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Many frameworks have been proposed for the management of uncertainty in logic programs (see, *e.g.* [1] for a list of references). Roughly, they can be classified into *annotation based* (AB) and *implication based* (IB) approaches. In the AB approach, a rule has form $A : f(\beta_1, \dots, \beta_n) \leftarrow B_1 : \beta_1, \dots, B_n : \beta_n$, asserting “the certainty of atom A is at least (or is in) $f(\beta_1, \dots, \beta_n)$, whenever the certainty of atom B_i is at least (or is in) β_i , $1 \leq i \leq n$ ” (f is an n -ary computable function and β_i is either a constant or a variable ranging over an certainty domain). In the (IB) approach, a rule has form $A \stackrel{\alpha}{\leftarrow} B_1, \dots, B_n$. Computationally, given an assignment v of certainties to the B_i s, the certainty of A is computed by taking the “conjunction” of the certainties $v(B_i)$ and then somehow “propagating” it to the rule head. While the way implication is treated in the AB approach is closer to classical logic, the way rules are fired in the IB approach are more intuitive. Broadly, the IB is considered easier to use and more amenable for efficient implementation. Anyway, a common feature of both approaches is that the assumption made about the atoms whose logical values cannot be inferred is *equal* for all atoms: in the AB approach the *Open World Assumption* (OWA) is used (the default truth value of any atom is *unknown*), while in the IB approach this default value is the *bottom element* of a truth lattice, *e.g.* *false*. We believe that we should be able to associate to a logic program a semantics based on any given hypothesis, which represents our default or assumed knowledge. To this end, we extended, by providing syntax and a fixpoint semantics, the *parametric* IB framework [1] (an unifying umbrella for IB frameworks) along two directions: (i) we introduced *non-monotonic negation* into the programs; and (ii) an atom’s truth may by default be either $\top$ (*e.g.* *true*), $\perp$ (*e.g.* *false*) or be *unknown*. A rule is of the form $r : A \stackrel{\alpha_r}{\leftarrow} B_1, \dots, B_n, \neg C_1, \dots, \neg C_m; \langle f_d, f_p, f_c \rangle$, in which f_d is a disjunction function associated with the predicate symbol A and, f_c and f_p are respectively a conjunction and a propagation functions associated with r . The intention is that the conjunction function determines the truth value of the conjunction of $B_1, \dots, B_n, \neg C_1, \dots, \neg C_m$, the propagation function determines how to “propagate” the truth value resulting from the evaluation of the body to the head, by taking into account the certainty α_r associated to the rule r , while the disjunction function dictates how to combine the certainties in case an atom is head of several rules. A *default assumption* is a partial function $H : B_P \rightarrow \{\top, \perp\}$, where B_P is the Herbrand base of a program P and $\top$ and $\perp$ are the top and bottom elements of a certainty lattice, respectively.

[1] Laks V.S. Lakshmanan and Nematollaah Shiri. A parametric approach to deductive databases with uncertainty. *IEEE Transactions on Knowledge and Data Engineering*, 13(4):554–570, 2001.

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